NEXUS ANALYTICS

Business Insights & Predictive Analytics Report

Synthetic Ghanaian mobile-money transaction-risk demonstration · 10 November 2026

1. Executive summary

This report analyses **30,000** reproducibly generated synthetic transaction records spanning 2026-01-01 to 2026-10-07. The dataset records a total transaction value of **GHS 1,628,477.48**, an average transaction value of **GHS 54.28**, and **622** records carrying a synthetic fraud label (2.07% of records). These labels are generated by an artificial risk mechanism and are not verified cases.

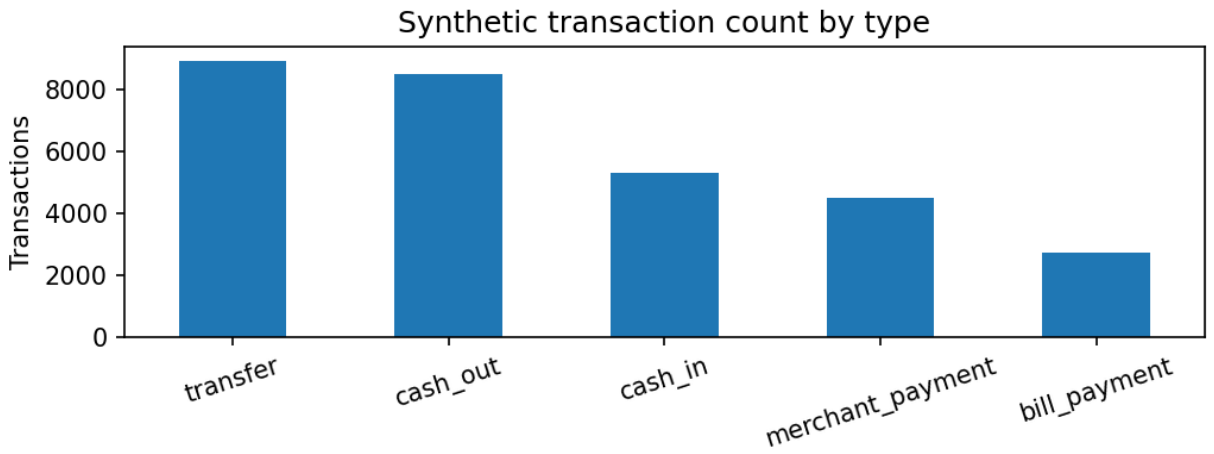
The SQL and dashboard layers describe transaction activity and data quality. A Random Forest model provides a reproducible classification example. Its ranking metrics exceed a DummyClassifier baseline, but precision remains low; outputs should be treated as review signals only.

2. Key performance indicators

Indicator	Observed value
Transaction records	30,000
Total transaction value	GHS 1,628,477.48
Average transaction value	GHS 54.28
Synthetic labelled-fraud records	622
Synthetic labelled-fraud share	2.07%
Distinct customers represented	4,989
Distinct agents represented	200

3. Transaction mix

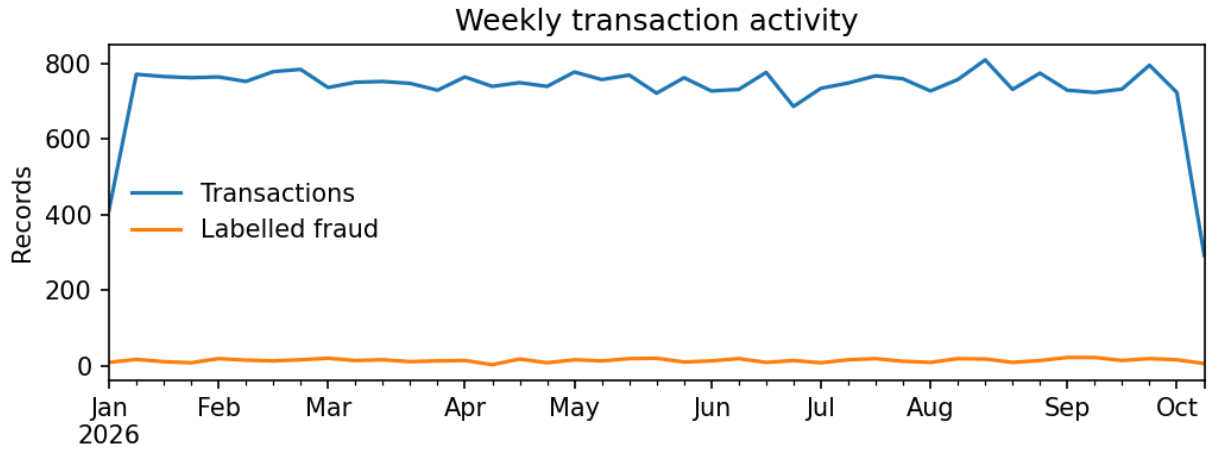
Transfer and cash-out records account for the largest transaction counts in the generated dataset. The chart below shows counts by transaction type; values are synthetic and should not be interpreted as a market-share estimate.



Transaction type	Count	Total value (GHS)
Transfer	8,946	498,238.74
Cash Out	8,499	445,992.97
Cash In	5,295	293,408.98
Merchant Payment	4,518	244,197.08
Bill Payment	2,742	146,639.71

4. Time pattern

The weekly chart summarises transaction volume and the number of records marked with the synthetic fraud label. Weekly changes reflect the generator's random timestamp allocation and should not be interpreted as actual seasonal or operational patterns.



5. Data quality observations

The generated dataset intentionally contains **45** missing customer identifiers and **20** channel values with trailing whitespace. The current dataset has no duplicate transaction identifiers, negative amounts or unparseable timestamps. These imperfections provide examples for SQL and application-level data-quality checks.

6. Machine-learning evaluation

Metric	Random Forest / selected threshold	Dummy baseline
ROC AUC	0.6268	0.5000
Average precision	0.0320	0.0207
Selected threshold	0.1267	—
Precision	4.08%	—
Recall	26.45%	—
F1	0.0707	—
Predicted-positive rate	13.40%	—

The model's ROC AUC is approximately 0.627 and average precision is approximately 0.032. At the holdout-F1-selected threshold of approximately 0.127, recall is approximately 26.5% but precision is approximately 4.1%. This means most records flagged at this threshold are not positive in the holdout labels. Threshold selection based only on F1 is not a substitute for a cost-based operating policy, calibration, time-based validation or independent testing.

7. Business implications and recommended next steps

• Use SQL aggregates and dashboard charts to prioritise further investigation of transaction types, channels and periods with higher activity. Do not infer risk from volume alone.
• Treat the model output as a queueing or review signal only. Combine it with additional evidence and human investigation before any customer-impacting action.
• Improve feature quality and model discrimination, test precision-recall trade-offs against review capacity, calibrate scores, and validate on time-separated and independently sourced data.
• In a production deployment, monitor feature drift, label delay, precision/recall by segment, false-positive burden, API latency and model version changes.

8. Scope and limitations

All figures in this report are calculated from synthetic data. No real customers, accounts or provider records are represented. The artificial label-generation process is not calibrated to a real operator. AI-generated management commentary is not included in this static report; the application can generate a contextual Snwolley report when a fresh API credential and live provider connectivity are configured. Model scores are not calibrated probabilities and must not be used as standalone fraud decisions.